remote sensing

Article

Asymmetric Frequency-Decoupled Network for Robust Visible–Infrared Fire Detection

Hongkai Chen [1], Hongtu Cai [2], Rong Sun [1], Xiumei Chen [3,*], Xin Chen [1] and Xiaoxing Zhu [1]

[1] China Tower Corporation Limited, Fujian Province Branch, Fuzhou 350018, China; chenhk@chinatowercom.cn (H.C.); sunrong@chinatowercom.cn (R.S.); chenxin6@chinatowercom.cn (X.C.); zhuxx@chinatowercom.cn (X.Z.)
[2] Academy of Digital China (Fujian), Fuzhou University, Fuzhou 350108, China; 245527028@fzu.edu.cn
[3] College of Physics and Information Engineering, Fuzhou University, Fuzhou 350108, China
* Correspondence: chen-xiumei@fzu.edu.cn

Highlights

What are the main findings?

- Infrared low-frequency features usually have stronger local contrast in thermal target regions. They can serve as reliability-related cues to guide the adaptive aggregation of visible low-frequency semantics.
- Smoke occlusion often causes low responses in visible high-frequency features. These low-response features can be converted into an occlusion-related soft gate through negative mapping, which adjusts the complementarity between visible and infrared high-frequency structural information.

What are the implications of the main findings?

- The results show that this asymmetric frequency-decoupled framework effectively improves the accuracy of visible–infrared fire detection.
- AFDNet provides a more targeted frequency-domain modeling strategy for visible–infrared fire detection under smoke occlusion and cross-modal contamination.

Abstract

Wildfires are highly destructive natural disasters posing serious threats to ecosystems. Visible–infrared fusion is an effective paradigm for robust fire detection in complex scenarios. However, existing spatial-domain fusion methods inevitably suffer from cross-modal contamination: the strong thermal radiation of flames dilutes visible textures, while dense visible smoke suppresses infrared targets. To circumvent this bottleneck, we reveal the frequency-domain asymmetry of fire features and propose an Asymmetric Frequency-Decoupled Network (AFDNet). By shifting from symmetric spatial fusion to asymmetric frequency decoupling, AFDNet effectively isolates modal conflicts, enabling targeted feature enhancement without mutual interference. Specifically, a Modality-Specific Frequency Decoupling (MFD) module first employs the Discrete Wavelet Transform to decompose features into low- and high-frequency sub-bands, breaking spatial entanglement. Subsequently, for low-frequency energy, a Thermal-Guided Low-Frequency Aggregation (TLA) module leverages infrared local contrast as a physical prior to guide fusion, ensuring precise thermal localization while preserving visible scene semantics. For high-frequency details, a Smoke-Masked High-Frequency Restoration (SHR) module maps smoke-induced visible high-frequency collapse into a soft reliability gate. This gate introduces infrared details to compensate for smoke-weakened structural cues. Extensive experiments on the RGBT-3M dataset demonstrate that AFDNet achieves state-of-the-art performance, outperforming the

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second-best method by 3.37% in mAP, while exhibiting exceptional robustness in complex wildfire environments.

Keywords: fire detection; visible–infrared fusion; frequency decoupling; asymmetric network; smoke occlusion; thermal radiation

1. Introduction

Wildfires represent one of the most destructive natural hazards, necessitating robust early detection systems to mitigate their irreversible impacts [1,2]. In the field of disaster perception, deep learning-based visual recognition methods have been extensively studied and show powerful detection capabilities. However, as shown in Figure 1, visible cameras become ineffective in dense smoke, while infrared sensors are insensitive to smoke. Consequently, visible–infrared (VIS-IR) cross-modal detection has emerged as a mainstream paradigm for fire detection [3,4]. This approach intrinsically leverages the complementary strengths of dual sensors: visible (VIS) cameras provide essential texture details and scene semantics, whereas infrared (IR) sensors penetrate dense smoke to accurately localize thermal radiation targets. By integrating these modalities, VIS-IR systems effectively overcome the visual degradation of VIS sensors in smoke [5–7] and the lack of contextual awareness in IR imagery [8,9], ensuring accurate fire detection.

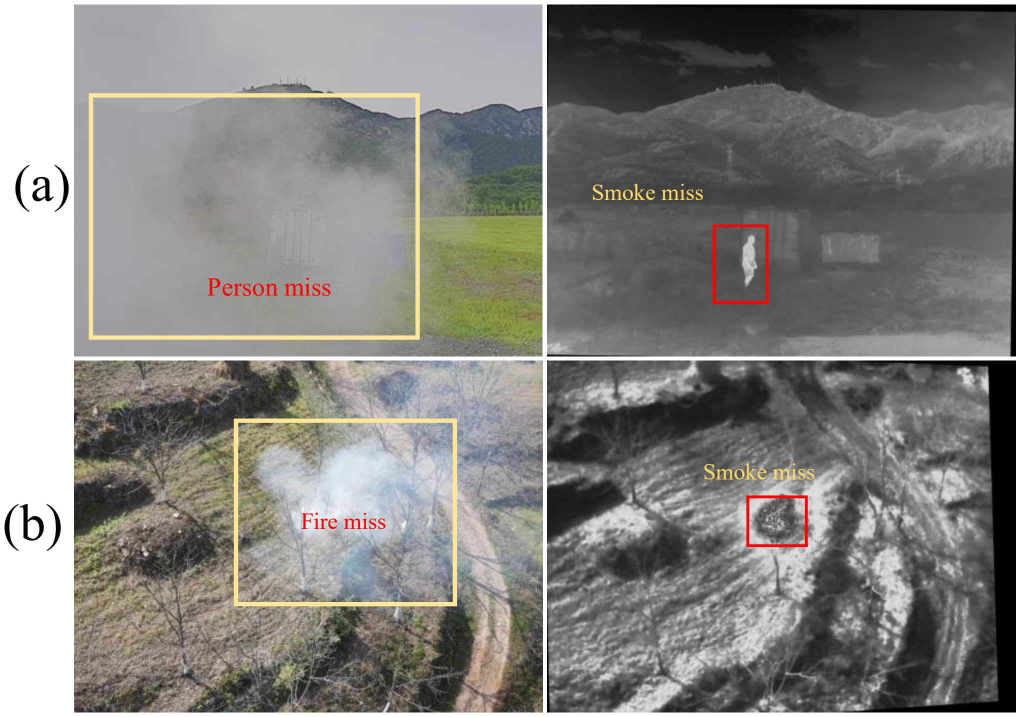

Figure 1. Limitations of single-modality fire detection: visible cameras are susceptible to smoke occlusion, while infrared cameras are insensitive to smoke. (**a**) From the M3FD dataset: smoke is clearly visible in the visible image, but personnel beneath the smoke are undetectable; the infrared image reveals the personnel targets while showing no response to the smoke region. (**b**) From the RGBT-3M dataset: smoke is clearly visible in the visible image, but fire sources beneath the smoke are undetectable; the infrared image clearly reveals the fire sources while showing no response to the smoke region.

Current VIS-IR cross-modal detection architectures generally fall into two dominant paradigms: two-stage and single-stage frameworks [10–14]. Two-stage methods adopt a cascaded "fusion-then-detection" pipeline [15,16], where modalities are initially integrated at the pixel level to generate a fused image, which is subsequently fed into a standard detector. However, treating image fusion and object detection as isolated sub-tasks inherently leads to optimization misalignment [17]. Specifically, pixel-level fusion prioritizes overall visual fidelity, which often dilutes critical semantic features necessary for downstream detection. In extreme wildfire scenarios with severe occlusions, this semantic degradation drastically exacerbates false positives and missed detections. To circumvent these architectural bottlenecks, single-stage methods have gained prominence by integrating feature-level fusion directly into the detection network [18,19]. By establishing cross-modal feature interactions within a unified end-to-end learning framework, these approaches dynamically extract and align complementary representations, significantly enhancing detection efficacy in complex environments [20].

Despite the inherent advantages of single-stage architectures, prevailing spatial-domain fusion frameworks struggle to reconcile the extreme physical disparities intrinsic to wildfire scenarios. Specifically, flames emit intense thermal radiation dominating the infrared spectrum, whereas dense smoke generates widespread, low-contrast occlusions in the visible domain [21,22]. When directly fused in the spatial domain, these stark differences inevitably trigger a dual-conflict mechanism of cross-modal contamination. Paradoxically, the widespread homogeneous smoke pixels in the visible spectrum tend to dilute the salient thermal signatures of infrared targets, elevating the risk of missed flame detections. Concurrently, the inherently smooth and low-frequency nature of infrared signals suppresses the discriminative high-frequency texture details of the visible spectrum, stripping targets of fine-grained structural semantics [23]. Consequently, breaking this spatial entanglement to preserve modality-specific advantages without mutual interference remains a formidable challenge.

To circumvent the inherent limitations of spatial-domain fusion, we shift our perspective to the frequency domain. As shown in Figure 2, by analyzing the shallow-level feature representations of typical fire scenarios, we uncover a profound physical asymmetry between the visible and infrared modalities across distinct frequency bands. Quantitative validation of this frequency-domain asymmetry is further provided in Appendix A. Specifically, in the low-frequency spectrum, the thermal radiation from flames and human bodies exhibits a highly concentrated energy distribution in the infrared modality. This thermal dominance provides a highly reliable, interference-free localization prior for anchoring targets in complex environments. Conversely, in the high-frequency sub-bands, visible images subjected to smoke occlusion suffer from severe texture loss, manifesting as homogeneous regions of collapsed responses. Rather than treating this high-frequency degradation as a mere informational deficit, we creatively reconceptualize it as a robust discriminative prior. This collapsed response serves as a natural occlusion mask, explicitly guiding the infrared high-frequency components to reconstruct the missing geometrical contours of obscured targets. This shift to the frequency domain is grounded in clear physical mechanisms. In local regions containing thermal targets, the strong concentration of infrared thermal radiation produces a local contrast that far exceeds that of visible light in the low-frequency subband. In smoke-occluded regions, smoke-particle scattering, which can be described by Mie scattering theory when particle sizes are comparable to visible wavelengths, tends to homogenize local spatial gradients and suppress edge and texture information, thereby leading to visible high-frequency signal collapse [24,25].

Driven by these physical insights, this paper proposes a novel single-stage Asymmetric Frequency-Decoupled Network (AFDNet) for robust visible–infrared fire detection. Deviating from conventional methods that heavily rely on spatial feature concatenation or weighted fusion, AFDNet shifts the cross-modal interaction to the frequency domain, establishing an integrated "decoupling-processing-reconstruction" pipeline. The architecture is built upon three core components. First, a Modality-Specific Frequency Decoupling (MFD) module fundamentally bypasses spatial information aliasing by employing wavelet transforms. This operation disentangles the entangled multimodal inputs into distinct low-frequency semantic representations and high-frequency structural details. Subsequently, a parallel asymmetric processing strategy is executed. For the low-frequency pathway, a Thermal-Guided Low-Frequency Aggregation (TLA) module leverages the saliency and local contrast of infrared thermal radiation as reliability-related cues for adaptive fusion. This encourages the model to rely more on thermal cues when they are reliable, while incorporating visible semantics for auxiliary discrimination. Conversely, for the high-frequency pathway, a Smoke-Masked High-Frequency Restoration (SHR) module maps the collapsed visible responses in smoke-obscured regions into a soft occlusion-related reliability gate. This gate guides the incorporation of infrared high-frequency features to supplement smoke-weakened structural cues. Ultimately, an inverse wavelet transformation maps these asymmetrically enhanced components back to the spatial domain, yielding highly discriminative fused representations that synthesize infrared smoke-penetration capabilities with visible structural fidelity.

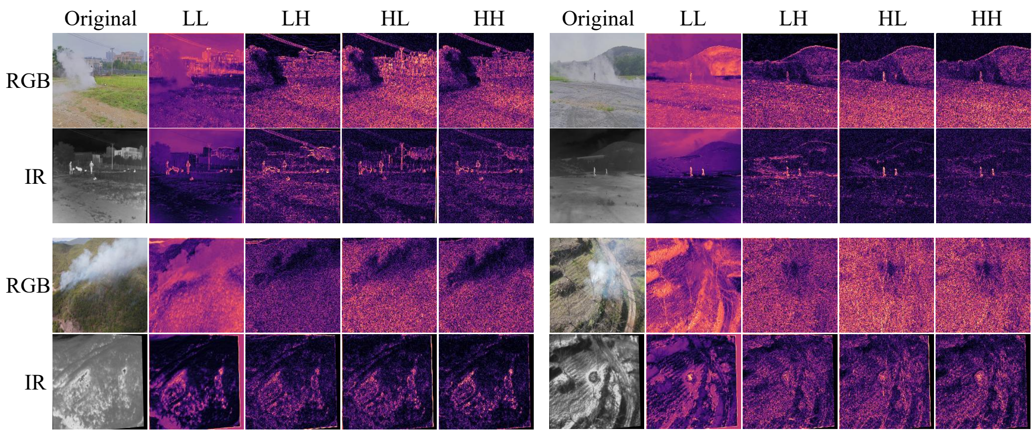

Figure 2. Original images and shallow wavelet transform outputs (LL, LH, HL, HH) of visible and infrared features under different fire scenarios.

The main contributions of this work are summarized as follows:

(1) A novel physical perspective based on frequency-domain asymmetry is introduced, which creatively reconceptualizes the high-frequency degradation of visible smoke as an occlusion-related reliability cue to alleviate cross-modal contamination.
(2) An Asymmetric Frequency-Decoupled Network (AFDNet) is proposed, which explicitly isolates modal conflicts via the frequency decoupling module (MFD) and independently enhances modality-specific dominant features through the carefully designed TLA and SHR modules.
(3) Extensive experiments demonstrate that the proposed method not only achieves state-of-the-art detection performance on the challenging RGBT-3M fire dataset but also exhibits robust generalization capabilities on the general-purpose M3FD cross-modal benchmark.

2. Related Works

2.1. Visible–Infrared Fusion Detection

Multimodal fusion detection aims to improve detection accuracy by leveraging the complementary nature of cross-modal information [26–28]. In general, existing visible–infrared fusion detection methods can be divided into two-stage and single-stage approaches [13,14]. The first category focuses on designing image fusion strategies to generate fused images that are rich in key information, after which a single-modal detector is used to complete the object detection task. Under this paradigm, researchers have proposed a variety of methods with distinct insights. For example, Liu et al. [29] proposed TarDAL, which builds a cooperative game between fusion and detection through two-level optimization, enabling task-driven preservation of object information. To overcome the limitations of fixed fusion rules in adapting to complex environments, Tang et al. [30] designed PIAFusion, which significantly improves the ability to capture salient objects under all-weather and extreme lighting conditions. As research attention shifted toward semantic-level interaction and robustness, Yi et al. [31] proposed Text-IF, which achieves perception and understanding of source image degradation by introducing text-based semantic guidance. Zhang et al. [32] built the DCEvo framework based on evolutionary learning, aiming to achieve the co-evolution of fusion quality and perceptual performance through cross-dimensional feature enhancement. To address the bottleneck of traditional models in deep semantic alignment, Wang et al. [33] introduced MTG-Fusion, which uses large-model-driven multi-text guidance to achieve dynamic alignment of fusion strategies.

However, two-stage methods treat fusion and detection as two separate optimization tasks. This means the optimal outcome for image fusion does not always align with the requirements of object detection, which limits overall detection performance. The second category attempts to integrate image fusion and object detection into a unified model. Feature-level fusion strategies have become the mainstream direction in single-stage methods, as they are better at exploring complex cross-modal relationships [34–36]. For example, Liu et al. [37] adopted a weight-sharing dual-stream network as the backbone, establishing a standard paradigm that combines a dual-stream architecture with intermediate-layer fusion. Zhao et al. [38] proposed RSDet, which filters out irrelevant cross-modal interference through a "coarse-to-fine" fusion strategy, enabling more precise selection of discriminative features. As research shifted toward deeper coupling between fusion and detection, Wang et al. [39] explored joint optimization strategies. By designing a cross-modal kernel interaction loss to guide the generation of dynamic convolutional kernels, they addressed the problem of insufficient feature extraction in traditional methods. Kim et al. [40] proposed a causal modal reuse framework from a causal reasoning perspective, aiming to reduce the model's false reliance on a single modality. Beyond handling external interference, managing information imbalance and redundancy across modalities has also become an important research direction. In addition, works such as FCRPN [41], CFT [42], and ICAFusion [43] have each presented improved solutions, further confirming the effectiveness of feature-level fusion in complex scenarios.

Furthermore, the study by Liu et al. [44] reveals that there are fundamental differences in physical imaging between the high-frequency details of the visible modality and the low-frequency contours of the infrared modality. Without processing tailored to frequency-domain characteristics, direct spatial fusion can lead to cross-modal contamination, which suppresses the saliency of true targets. To address this, the wavelet transform, which decomposes images into different frequency subbands while preserving spatial information, has shown strong potential in fields such as object detection [45], fault diagnosis [46,47], and signal detection [48,49]. In cross-modal detection research, Zhu et al. [50] proposed WaveMamba, which uses the Discrete Wavelet Transform (DWT) to integrate the unique frequency features of visible and infrared modalities, confirming that complementary frequency-domain information plays an important role in improving detection robustness. Building on this, this paper further develops the frequency-domain processing strategy by applying wavelet transforms to decompose visible and infrared features at multiple scales, and designs separate fusion schemes for high-frequency and low-frequency components.

2.2. Visible and Infrared Fire Detection

The technological advances in cross-modal object detection described above have laid a solid foundation for fire detection. However, unlike general objects with fixed shapes such as pedestrians and vehicles, flames and smoke are highly diffuse and tend to cause heavy occlusion [51,52]. These unique imaging characteristics make cross-modal contamination especially likely when processing different types of information in fire scenes.

To address the specific needs of fire scenarios, researchers have explored various fusion strategies. Chen et al. [53] systematically compared the performance of early fusion and late fusion strategies in UAV fire monitoring, providing early evidence that multimodal data can improve fire detection accuracy. To further reduce false alarms caused by environmental noise, Rask et al. [54] explored decision-level fusion. Through an independent validation strategy, they effectively suppressed illumination interference and fire-like objects that appear in single modalities, improving the overall robustness of the system.

As research moved toward deeper feature-level interaction, Goswami et al. [55] used a dual-stream model based on MobileNetV2 to show that feature-level interaction can both use infrared features to compensate for missed detections caused by smoke occlusion and use visible textures to suppress hot-spot false positives in infrared images. With the rise of Transformer architectures, Wang et al. [56] applied a cross-attention mechanism to give the network dynamic modal selection ability. When visible features are obscured by smoke, the model can automatically shift its focus to the corresponding infrared thermal radiation features, achieving more precise cross-modal feature alignment and fusion.

3. Method

3.1. Overall Architecture

Figure 3 illustrates the overall architecture of AFDNet. Given a pair of visible and infrared images, AFDNet uses a dual-stream YOLO11 backbone network to extract multi-scale feature representations for both modalities. At feature levels P3, P4, and P5, AFDNet outputs the visible feature maps F^i_{VIS} and infrared feature maps F^i_{IR} at each scale ($i \in \{3, 4, 5\}$). These feature maps form a multi-scale feature stream from low-level details to high-level semantics, and this stream is then passed to the cross-modal fusion module.

At each scale, AFDNet follows a three-step processing workflow. First, the Modality-Specific Frequency Decoupling mechanism (MFD) applies 2D-DWT to decompose both F^i_{VIS} and F^i_{IR} into low-frequency and high-frequency subbands. This step separates the entangled modality information, including heat source energy and smoke texture, from the spatial domain into decoupled frequency components. Next, for the low-frequency subbands, the Thermal-Guided Low-Frequency Aggregation (TLA) module uses the local contrast of the infrared modality as an adaptive weight to perform asymmetric aggregation. In this process, infrared features provide reliable localization cues, and visible features provide supplementary scene semantics, which prevents the visible background from weakening the infrared thermal radiation signals. At the same time, for the high-frequency subbands, the Smoke-Masked High-Frequency Restoration (SHR) module handles smoke-covered regions by converting collapsed dark clusters in the visible high-frequency data into soft occlusion-prior masks through negative response mapping. These masks then guide the infrared high-frequency features to supplement the missing structural cues in occluded regions. The detailed structure and mathematical formulation of each module are presented in Sections 3.2, 3.3 and 3.4, respectively.

After fusion, the low-frequency and high-frequency subbands processed by TLA and SHR are remapped back to the spatial domain through IDWT operations, producing a fused feature map with the same dimensions as the original input. To compensate for any loss of modal details during frequency-domain processing, the reconstructed feature is combined with the original features through a residual connection with a scaling factor of 0.2, forming the final multi-scale fused feature F^i_{final}. The fused features are then fed into the YOLO11 Neck network for multi-scale aggregation, and the detection head outputs the final object categories and bounding box locations.

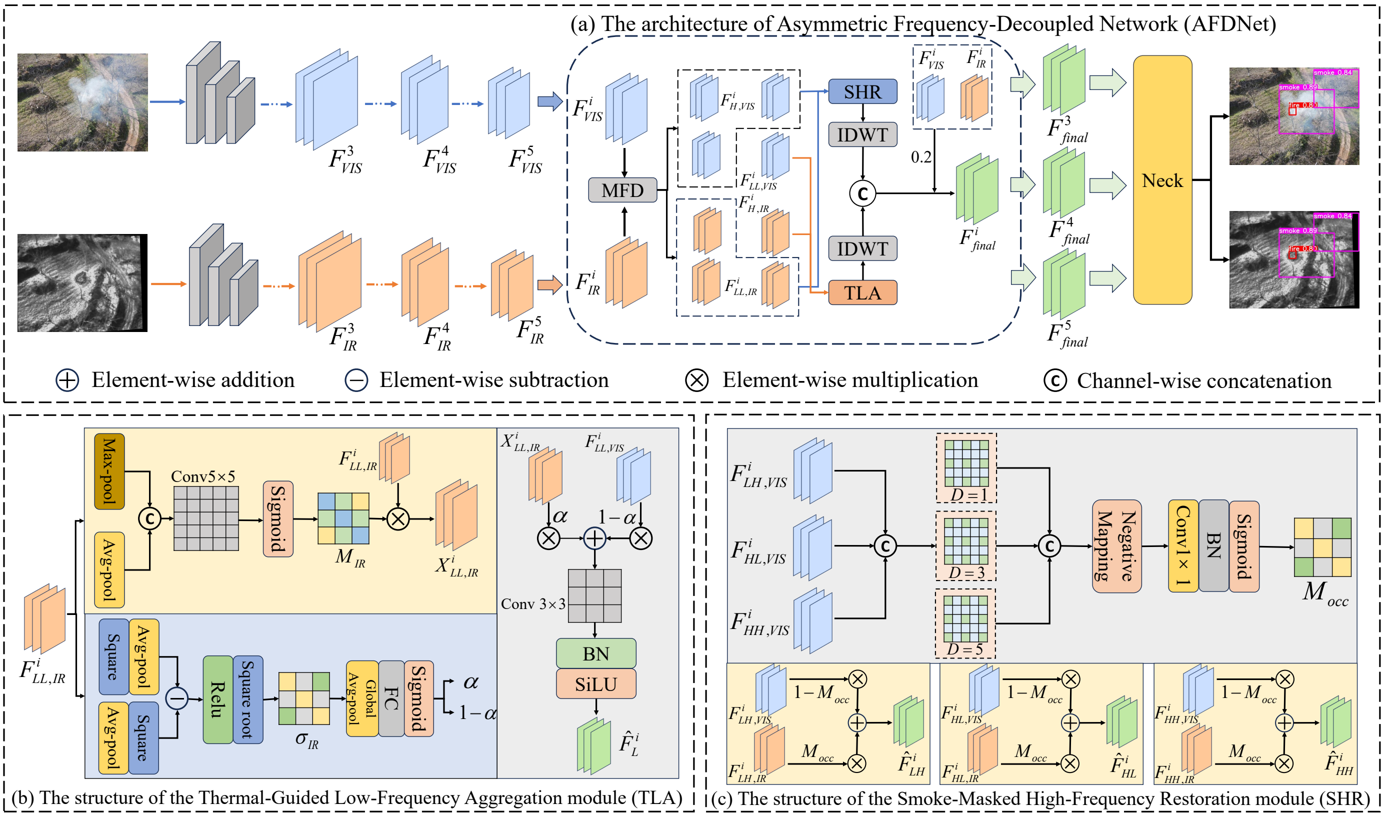

Figure 3. Schematic diagram of the overall architecture and core sub-modules of AFDNet. (**a**) Overall architecture of AFDNet: The network adopts a dual-stream backbone to extract multi-scale visible and infrared features. At feature levels P3, P4, and P5, features undergo MFD decoupling, TLA low-frequency aggregation, and SHR high-frequency restoration. The fused features are then sent to the Neck and detection head through residual connections to output the final detection results. (**b**) Detailed structure of the Thermal-Guided Low-Frequency Aggregation (TLA) module. (**c**) Detailed structure of the Smoke-Masked High-Frequency Restoration (SHR) module. In this figure, $+$, $-$, and $\times$ denote element-wise addition, subtraction, and multiplication, respectively, while C denotes channel-wise concatenation. The "Negative Mapping" block in (**c**) denotes the unary sign-inversion operation applied to X_{feat}, corresponding to the $-X_{feat}$ term in Equation (8).

3.2. Modality-Specific Frequency Decoupling

In fire scenarios, directly fusing infrared thermal radiation and visible smoke signals in the spatial domain can cause cross-modal contamination. To address this problem, this paper introduces the Modality-Specific Frequency Decoupling mechanism (MFD), which transfers cross-modal features from the spatial domain to the frequency domain. This step separates low-frequency energy from high-frequency details and provides clear inputs for the subsequent asymmetric processing modules.

Specifically, for each input feature map $F^i \in \mathbb{R}^{C\times H\times W}$ ($i \in \{3,4,5\}$), MFD applies a two-dimensional discrete wavelet transform (2D-DWT) with a Haar wavelet basis to decompose it into one low-frequency subband and a set of three directional high-frequency subbands:

$$\mathrm{DWT}(F^i) \to \{F^i_{LL}, F^i_{H}\}, \quad F^i_H = \{F^i_{LH}, F^i_{HL}, F^i_{HH}\}, \quad F^i_{LL}, F^i_{LH}, F^i_{HL}, F^i_{HH} \in \mathbb{R}^{C\times\frac{H}{2}\times\frac{W}{2}} \tag{1}$$

In this process, F^i_{LL} captures the global energy and low-frequency contours of the features, while F^i_H (which consists of F^i_{LH}, F^i_{HL}, and F^i_{HH}) corresponds to high-frequency details in the horizontal, vertical, and diagonal directions, respectively. This decomposition is applied independently to the visible and infrared features, producing $F^i_{LL,VIS}$, $F^i_{H,VIS}$, $F^i_{LL,IR}$, and $F^i_{H,IR}$ as inputs for the subsequent TLA and SHR modules.

To preserve the spatial-frequency localization essential for fire target detection, this paper adopts the 2D Discrete Wavelet Transform (2D-DWT) rather than global spectral methods to separate the features. After fusion, the frequency-domain subbands processed by TLA and SHR are remapped back to the spatial domain through the inverse discrete wavelet transform (IDWT), restoring the fused feature map $F^i_{\text{final}} \in \mathbb{R}^{C\times H\times W}$ with the same dimensions as the original input for subsequent target localization and classification.

3.3. Thermal-Guided Low-Frequency Aggregation

In the low-frequency subband, the thermal energy of the human body and flames is usually concentrated in the infrared modality $F^i_{LL,IR}$, producing salient low-frequency responses and often providing reliable cues for target localization. In contrast, the visible low-frequency modality $F^i_{LL,VIS}$ may suffer from reduced target-background contrast due to smoke coverage and lighting interference, while still retaining complementary scene semantics. If a symmetric fusion strategy is applied, the homogeneous background of the visible low-frequency band may dilute the peak response of infrared thermal targets, weakening the model's ability to locate high-temperature fire points and people in danger. To tackle this challenge, this paper proposes the Thermal-Guided Low-Frequency Aggregation (TLA) module, which uses infrared local contrast as an adaptive weighting cue to establish an asymmetric aggregation mechanism where infrared features guide target localization and visible features provide supplementary scene semantics.

To fully exploit the localization advantages of the infrared modality, TLA first processes the infrared low-frequency features through a saliency enhancement branch. This branch applies max-pooling and average-pooling in parallel along the channel dimension. Max-pooling captures the peak radiation intensity of thermal targets, while average-pooling preserves the overall contour distribution of these targets. The two pooling outputs are concatenated and then passed through a convolutional layer and a Sigmoid activation function to generate a pixel-wise spatial attention mask M_{IR}:

$$M_{IR} = \sigma\Big(\mathrm{Conv}_{5\times5}\Big(\Big[\mathrm{AvgPool}(F^i_{LL,IR}); \mathrm{MaxPool}(F^i_{LL,IR})\Big]\Big)\Big), \tag{2}$$

where σ denotes the Sigmoid activation function, $\mathrm{Conv}_{5\times5}$ represents a 5×5 convolutional layer, and $[\cdot;\cdot]$ indicates concatenation along the channel dimension. Subsequently, this mask is multiplied with the infrared low-frequency features to produce the attention-enhanced feature $X^i_{LL,IR}$:

$$X^i_{LL,IR} = F^i_{LL,IR} \otimes M_{IR}, \tag{3}$$

where the symbol $\otimes$ represents element-wise multiplication.

In addition, TLA includes a modality reliability-aware branch that uses local contrast as a reliability-related cue for infrared features in the current scene. Since thermal target regions generally exhibit stronger local fluctuations than homogeneous backgrounds in the infrared low-frequency subband, TLA computes the standard deviation map σ_{IR} of the infrared features within a local window as a contrast metric:

$$\sigma_{IR} = \sqrt{E\left[(F^i_{LL,IR})^2\right] - \left[E(F^i_{LL,IR})\right]^2}, \tag{4}$$

Here, $E[\cdot]$ indicates the mathematical expectation within the local window. This map is then passed through global average pooling and fully connected layers to produce the modal fusion attention weight α, which serves as a learnable weighting factor for modulating the relative contributions of the two modalities during aggregation:

$$\alpha = \sigma(\mathrm{FC}(\mathrm{GAP}(\sigma_{IR}))), \tag{5}$$

where GAP represents the Global Average Pooling operation and FC denotes the fully connected layer. Since the weight α is adaptively generated from the local infrared contrast representation, it provides a data-driven indication of the relative reliability of infrared cues in the current scene. In this context, α functions as a learnable gate for modality-specific reliability estimation, rather than a manually defined infrared-dominant coefficient. In challenging scenarios where infrared contrast may be degraded, such as high-ambient-heat environments, this reliability-aware branch allows the network to reduce its reliance on infrared cues and incorporate more visible low-frequency information when necessary. This design helps mitigate the risk of over-relying on unreliable thermal signals and enables the model to adapt its use of infrared information in a data-driven manner, rather than relying on manually defined fixed fusion ratios.

In the final aggregation stage, the attention-enhanced infrared feature $X^i_{LL,IR}$ is fused with the visible low-frequency feature $F^i_{LL,VIS}$ through asymmetric weighted fusion. The infrared feature is weighted by the learned coefficient α, while the visible feature is weighted by $1-\alpha$. Therefore, the module does not use a fixed infrared-dominant rule. Instead, it provides a learnable mechanism for adjusting the low-frequency fusion ratio according to the local contrast representation of infrared features. By incorporating the $(1-\alpha)$ component, visible semantics can participate in auxiliary discrimination, helping the model maintain categorical perception when infrared localization cues become less reliable. The fused result is then smoothed by a 3×3 convolutional block to reduce boundary artifacts introduced by frequency-domain processing:

$$\hat{F}^i_L = \mathrm{SiLU}\Big(\mathrm{BN}\Big(\mathrm{Conv}_{3\times3}\Big(\alpha \cdot X^i_{LL,IR} + (1-\alpha)\cdot F^i_{LL,VIS}\Big)\Big)\Big). \tag{6}$$

Here, $\hat{F}^i_L$ is the output feature of the Thermal-Guided Low-Frequency Aggregation module, where SiLU indicates the Sigmoid Linear Unit activation function and BN represents Batch Normalization. This fusion mechanism is driven by infrared local contrast and built around asymmetric weighting. It encourages the model to make greater use of infrared thermal cues when they are reliable, while still allowing visible low-frequency semantics to support discrimination through the $(1-\alpha)$ term.

As discussed, the fusion weight α is learned from the local thermal contrast representation. When the thermal contrast between the target and the background becomes weak, such as in high-ambient-heat or desert environments, the reliability-aware branch is expected to help the model reduce its reliance on infrared cues and allow more visible low-frequency information to participate in the fusion. However, the current RGBT-3M dataset does not contain enough high-background-temperature scenes for a full validation. Thus, the robustness of TLA under extremely low thermal contrast remains a limitation of this work.

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3.4. Smoke-Masked High-Frequency Restoration

The collapse of visible high-frequency signals under smoke occlusion can be physically explained. Wildfire smoke particles typically have a median radius of 0.1 to 0.2 µm, which is on the same order of magnitude as the wavelength of visible light [24]. According to Mie scattering theory, strong scattering occurs under this condition [25]. Such scattering tends to homogenize the high-frequency components that carry edge and texture information. As a result, large low-response areas are often observed in the visible high-frequency subband of smoke-covered regions.

In the high-frequency subband, the scattering effect of smoke may cause visible high-frequency features to degrade into large, low-variance dark patches in occluded areas, resulting in local texture loss. Traditional fusion methods may treat these dark patches as invalid noise and suppress them. However, in fire scenes, their spatial distribution often overlaps with smoke-obscured regions and can provide a useful indication of potentially unreliable visible high-frequency responses. Although the infrared high-frequency subband may contain a cluttered background and weak target contours, it still retains useful structural information for smoke-penetrating target perception. Based on the idea of converting signal degradation into a reliability-related cue, this paper proposes the Smoke-Masked High-Frequency Restoration (SHR) module. Through negative response mapping, it converts collapsed dark patches in the visible high-frequency band into a soft occlusion-related reliability gate, allowing infrared high-frequency details to supplement the structural cues that may be weakened in smoke-obscured regions.

To estimate potentially unreliable low-texture regions in the visible high-frequency subband, SHR first aggregates the visible high-frequency subbands from three directions ($F^i_{LH,VIS}$, $F^i_{HL,VIS}$, and $F^i_{HH,VIS}$) into a combined high-frequency feature $\tilde{F}^i_{H,VIS}$. This feature is then passed through parallel multi-scale dilated convolutional layers to expand the receptive field and capture low-texture regions of varying sizes, producing the multi-scale aggregated feature X_{feat}:

$$X_{feat} = \mathrm{Cat}(\mathrm{DConv}_{d=1}(\tilde{F}^i_{H,VIS}), \mathrm{DConv}_{d=3}(\tilde{F}^i_{H,VIS}), \mathrm{DConv}_{d=5}(\tilde{F}^i_{H,VIS})), \tag{7}$$

where Cat denotes channel-wise concatenation, and DConv_d represents a dilated convolution with a dilation rate of d.

Subsequently, a negative response mapping is applied to the aggregated feature X_{feat}. In the visible high-frequency subband, smoke-obscured regions tend to lose texture and manifest as low-response dark patches. By applying a negative sign to the aggregated feature X_{feat}, we introduce an inverse response representation that provides the subsequent learnable transformation with a cue related to low visible high-frequency responses. This inverted representation is then passed through a 1×1 convolutional layer, Batch Normalization, and a Sigmoid activation function to generate the spatial occlusion-related reliability gate M_{occ}:

$$M_{occ} = \sigma(\mathrm{BN}(\mathrm{Conv}_{1\times1}(-X_{feat}))). \tag{8}$$

Here, $\mathrm{Conv}_{1\times1}$ indicates a 1×1 convolutional layer, and σ represents the Sigmoid function. It should be noted that the negative sign itself does not directly produce the final mask. Instead, the combination of negative response mapping, learnable 1×1 convolution, Batch Normalization, and Sigmoid activation allows the network to generate a soft gate for modulating the contribution of visible and infrared high-frequency features.

A high value of M_{occ} suggests that the visible high-frequency information in the corresponding region may be unreliable, which is often associated with smoke occlusion in fire scenes. In contrast, a low value of M_{occ} suggests that the visible high-frequency information is relatively reliable. This negative response mapping logic reinterprets low-

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response visible high-frequency regions as reliability-related cues, providing a pixel-level soft gating signal for subsequent cross-modal detail restoration.

It should be noted that low-value regions in the visible high-frequency subband are not solely caused by smoke. Other regions, such as dark sky areas, terrain shadows, and dense vegetation, may also show weak high-frequency responses. Therefore, M_{occ} is designed as a soft replacement gate for feature fusion, rather than a strictly supervised smoke probability map. A high value of M_{occ} suggests that the visible information may be unreliable, which prompts the network to introduce more infrared details. Although M_{occ} does not independently determine the final detection result, it can still affect the feature fusion process.

However, this design cannot completely eliminate the risks associated with ambiguous low-response regions. These non-smoke dark areas can still affect the fusion process and might increase the risk of false alarms under extreme conditions. The current RGBT-3M dataset does not contain enough of these extreme non-smoke scenes for a full quantitative study. Therefore, the performance of the SHR module in these boundary cases remains a limitation of this work. We plan to explore this further with more diverse visible–infrared fire datasets.

During the high-frequency detail restoration phase, SHR uses the occlusion-related reliability gate M_{occ} to perform pixel-level dynamic balancing between the infrared and visible high-frequency subbands. In regions where M_{occ} approaches 1, the fusion operator reduces the contribution of unreliable visible high-frequency responses and introduces more infrared high-frequency information to supplement structural cues. In regions where M_{occ} approaches 0, it preserves more texture details from the visible high-frequency subband:

$$\begin{cases} \hat{F}^i_{LH} = (1 - M_{occ}) \otimes F^i_{LH,VIS} + M_{occ} \otimes F^i_{LH,IR} \\ \hat{F}^i_{HL} = (1 - M_{occ}) \otimes F^i_{HL,VIS} + M_{occ} \otimes F^i_{HL,IR} \\ \hat{F}^i_{HH} = (1 - M_{occ}) \otimes F^i_{HH,VIS} + M_{occ} \otimes F^i_{HH,IR} \end{cases}. \tag{9}$$

The above fusion process is applied independently to the horizontal, vertical, and diagonal high-frequency subbands. It helps reduce the influence of unreliable visible high-frequency responses in potentially occluded regions and uses the infrared modality to supplement structural information for smoke-affected target perception.

3.5. Loss Function

AFDNet is trained end-to-end, and its loss function follows the standard detection loss design from YOLO11 [57], which consists of a classification loss $\mathcal{L}_{cls}$, a bounding box regression loss $\mathcal{L}_{box}$, and a distribution focal loss $\mathcal{L}_{dfl}$ [58,59]. The total loss is defined as the weighted sum of these three components:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{cls} + \lambda_2 \mathcal{L}_{box} + \lambda_3 \mathcal{L}_{dfl}, \tag{10}$$

where λ_1, λ_2, and λ_3 are the weighting coefficients for each loss term, using the default settings from YOLO11. Since the loss function design is not a core contribution of this paper, further details are omitted here.

4. Experiment

4.1. Dataset and Experimental Setup

RGBT-3M [60] contains 11,220 pairs of synchronized images captured by a drone-mounted dual-spectral camera, covering both visible and thermal infrared modalities. The data were collected from forested areas in Anhui, Yunnan, and Inner Mongolia, China, documenting real-world wildfires and field fire experiments. It captures the evolution of wildfires from multiple angles and scales, as well as scenes involving complex lighting conditions and vegetation occlusion. The dataset has been manually annotated for three categories: "smoke," "flame," and "people," with a total of 13,574, 11,315, and 5888 annotated instances, respectively. The dataset is split into a training set and a validation set in a 7:3 ratio to test the model's robustness in multimodal fusion and small-object detection tasks.

M3FD [29] contains 4200 pairs of synchronized images captured by a dual-spectral camera, covering both visible and thermal infrared modalities. It includes manual annotations for six categories: "people," "passenger cars," "buses," "motorcycles," "trucks," and "streetlights," and is one of the most widely used benchmark datasets in the field of dual-modal object detection. Although its annotation categories do not directly correspond to fire detection tasks, the dataset includes some scenes with smoke occlusion caused by fires. Therefore, we use M3FD as a generalization validation dataset to evaluate AFDNet's cross-scenario adaptability in non-fire dual-modal settings. The dataset is split into a training set and a validation set in a 4:1 ratio.

Experimental Setup. The proposed method is implemented using the PyTorch 1.13.0+cu117 framework and trained on a single NVIDIA GeForce RTX 4090 GPU. It extends the single-stage YOLO11 detector to a dual-stream architecture. This architecture extracts backbone features from each modality separately and then applies the proposed fusion method to generate multi-scale fused features for fire detection. The input image size is set to 640×640, and the batch size is set to 16. AFDNet is applied to fuse backbone features at three different scales (P3, P4, and P5 in Figure 3), which provides a good balance between performance and computational efficiency. Following the default hyperparameters of the YOLO series, training is performed for 250 epochs using an SGD optimizer, with an initial learning rate of 0.01, a momentum of 0.937, and a weight decay of 0.0005.

Metrics. This paper uses mean Average Precision (mAP) as the core evaluation metric. Specifically, mAP50 and mAP75 measure detection performance at IoU thresholds of 0.5 and 0.75, respectively, reflecting the model's ability under general detection and high-precision localization requirements. The overall mAP measures robustness by averaging precision across multiple IoU thresholds. Because the target categories are different in appearance, we also report per-category detection results for smoke, flame, and people in the comparative experiments.

4.2. Comparison with State-of-the-Art Methods

To comprehensively evaluate AFDNet, we compared it with several mainstream visible–infrared fusion detection methods on the RGBT-3M and M3FD datasets. The compared methods include classic two-stage approaches such as CDDFuse [61] and TarDAL [29], as well as single-stage methods such as ICAFusion [43], CFT [42], and GM-DETR [62]. To ensure a fair comparison, the two-stage methods were evaluated using the same detector, YOLO11, after image fusion. For the single-stage methods, we kept the original authors' default parameter settings but adjusted the model scale to match that of our method. Single-modality baselines trained on these datasets are also included for reference.

(1) Results on the RGBT-3M Dataset. Table 1 presents the detection performance of each method on the RGBT-3M dataset. Among all compared methods, AFDNet achieved the best performance across all metrics, with mAP50, mAP75, and mAP of 96.67%, 74.90%, and 66.94%, respectively. These results are 1.26%, 4.96%, and 3.37% higher than those of the second-best method, DCEvo [32]. The improvement in mAP75 is particularly notable. This suggests that AFDNet can achieve more precise target localization under high IoU thresholds. In terms of per-category performance, AFDNet achieved AP scores of 76.41%, 62.27%, and 62.13% for smoke, fire, and people, respectively, all of which are the best among all compared methods. The improvements over the second-best method reached 4.40% and 3.80% for the fire and people categories, respectively, demonstrating the advantage of the TLA module in localizing thermal radiation targets. The improvement in the smoke category also shows the effectiveness of the SHR module, which converts occlusion regions into prior masks to guide feature restoration.

Table 1. Performance comparison of different methods on the RGBT-3M dataset.

Methods	mAP50	mAP75	mAP	Smoke	Fire	People
IR	92.13	64.37	59.65	66.24	56.08	56.62
VIS	93.74	64.33	60.23	74.55	49.91	56.22
CDDFuse [61]	93.76	64.87	60.52	71.78	53.93	55.86
EI^2Det [13]	96.20	66.67	61.03	70.16	56.30	56.63
GM-DETR [62]	96.19	66.68	61.06	70.22	56.43	56.53
CFT [42]	96.17	67.64	61.56	70.68	56.91	57.08
U2Fusion [63]	94.14	66.88	61.77	72.39	56.24	56.69
ICAFusion [43]	96.34	67.81	62.17	71.64	57.23	57.64
MetaFusion [64]	94.98	67.14	62.39	74.60	55.34	57.22
TarDAL [29]	95.39	69.23	63.21	73.51	58.27	57.85
ReFusion [65]	95.43	69.34	63.40	74.54	57.63	58.03
DCEvo [32]	95.41	69.94	63.57	74.52	57.87	58.33
AFDNet (ours)	96.67	74.90	66.94	76.41	62.27	62.13

(2) Results on the M3FD Dataset. Table 2 presents the performance comparison of various methods on the M3FD dataset. M3FD covers six general object categories, including people, vehicles, and streetlights, and differs significantly from fire detection scenarios in terms of domain. Therefore, this experiment is mainly intended to evaluate AFDNet's cross-scenario generalization ability. AFDNet achieved mAP50, mAP75, and mAP scores of 84.79%, 63.28%, and 58.62%, respectively, representing improvements of 3.83%, 4.49%, and 4.19% over the second-best method, ReFusion [65], and ranked first among all compared methods. In terms of per-category performance, AFDNet achieved the best results in all six object categories. These results show that the frequency decoupling and asymmetric fusion mechanisms of AFDNet also transfer well to general dual-modal detection tasks.

Table 2. Performance comparison of different methods on the M3FD dataset.

Methods	mAP50	mAP75	mAP	People	Car	Bus	Lamp	Motor	Truck
IR	75.46	53.09	50.02	48.30	62.04	71.48	27.87	35.36	55.05
VIS	76.77	53.67	50.87	36.43	65.76	72.27	37.92	36.97	55.83
CDDFuse [61]	79.13	56.70	53.20	45.35	66.43	74.67	38.82	37.48	56.44
EI^2Det [13]	82.54	53.43	51.76	42.93	62.79	71.29	41.55	36.65	55.34
GM-DETR [62]	81.64	53.12	50.71	42.38	62.02	70.44	39.18	36.01	54.21
CFT [42]	81.72	53.43	50.50	42.57	61.81	70.50	39.28	35.32	53.56
U2Fusion [63]	80.16	57.53	53.71	47.29	66.89	73.40	38.80	38.33	57.54
ICAFusion [43]	82.31	53.61	51.31	42.83	62.69	70.25	42.21	35.54	54.37
MetaFusion [64]	79.16	55.45	52.45	44.85	66.10	73.85	37.81	35.06	57.02
TarDAL [29]	79.12	56.40	52.85	48.44	65.85	73.34	34.73	37.26	57.46
ReFusion [65]	80.96	58.79	54.43	46.45	67.42	72.87	41.37	39.65	58.79
DCEvo [32]	80.02	57.28	53.74	44.74	67.00	72.69	40.81	37.91	59.29
AFDNet (ours)	84.79	63.28	58.62	53.33	70.14	77.23	44.36	42.10	64.58

4.3. Ablation Experiments

To validate the effectiveness of each component in AFDNet, this section conducts an ablation analysis on the internal structures of the two core modules, TLA and SHR, using the RGBT-3M dataset, and systematically evaluates the overall contributions of all three components using the M3FD dataset. Comparative experiments are also conducted to verify the necessity of the low-frequency guidance strategy and multimodal input.

(1) Internal Ablation of the TLA Module. Table 3 presents a stepwise ablation of two key design elements in the TLA module. When the TLA module was completely removed, the mAP dropped from 66.94% to 66.12%, a decrease of 0.82%, demonstrating the necessity of low-frequency asymmetric aggregation for fire target localization. When only the saliency enhancement branch was removed, the mAP dropped to 66.44%, indicating that the parallel attention mechanism combining max-pooling and average-pooling makes a meaningful contribution to strengthening the peak response of infrared thermal targets. When only the reliability-aware branch was removed, the mAP dropped to 66.59%, showing that the adaptive weight generation mechanism based on local contrast effectively improves the model's ability to assess infrared signal quality across different scenarios. The best performance was achieved when both components were used together, confirming that the combination of saliency enhancement and reliability awareness is key to the effectiveness of the TLA module.

Table 3. Internal ablation study of TLA.

Methods	mAP50	mAP75	mAP
w/o TLA	96.22	73.03	66.12
w/o Saliency Enhancement	96.39	74.36	66.44
w/o Reliability Awareness	96.46	74.59	66.59
AFDNet (ours)	96.67	74.90	66.94

(2) Internal Ablation of the SHR Module. Table 4 presents stepwise ablation results for two key design elements of the SHR module. When the SHR module was completely removed, the mAP dropped from 66.94% to 66.27%, a decrease of 0.67%, confirming the important role of high-frequency smoke perception and restoration in complex occlusion scenarios. Among the subcomponents, removing the multi-scale dilated convolution aggregation reduced the mAP to 66.38%, indicating that multi-scale receptive fields are necessary for identifying smoke-covered regions of varying sizes. Removing the low-value inversion mechanism reduced the mAP to 66.39%. This variant reverses the negative sign in Equation (8), which degrades the module into a conventional enhancement of high-response regions, eliminating the directional semantic guidance toward occlusion regions. The resulting performance drop shows that the occlusion prior mask generated by negative response mapping is essential for guiding infrared high-frequency components to perform precise structural completion. This directly validates the design idea of converting visible high-frequency signal collapse into a useful prior. The best performance was achieved when both subcomponents were retained, further confirming that multi-scale perception and occlusion-aware prior generation work better together.

Table 4. Internal ablation study of SHR.

Methods	mAP50	mAP75	mAP
w/o SHR	96.27	73.40	66.27
w/o Multi-scale Aggregation	96.33	73.51	66.38
w/o Negative Response Mapping	96.39	74.47	66.39
AFDNet (ours)	96.67	74.90	66.94

(3) Comprehensive Ablation of Individual Components. Table 5 presents a systematic ablation of the three components, MFD, TLA, and SHR, on the M3FD dataset. On the dual-stream baseline without any components, mAP50, mAP75, and mAP were 80.94%, 59.67%, and 55.34%, respectively. After introducing MFD alone, these metrics improved to 81.42%, 60.37%, and 55.72%, respectively, corresponding to a modest mAP improvement of only 0.38%. When TLA was further added on top of MFD, the mAP increased from 55.72% to 58.31%, bringing an additional gain of 2.59%. When SHR was added on top of MFD, the mAP increased from 55.72% to 56.38%, with an additional gain of 0.66%. These results indicate that the main performance gain does not come from MFD alone, but from the subsequent asymmetric processing strategies, especially the low-frequency thermal-guided aggregation in TLA. When all three components were used together, mAP50, mAP75, and mAP reached their best values of 84.79%, 63.28%, and 58.62%, respectively, a further gain of 0.31% compared to using MFD and TLA alone. This confirms the complementary value of the SHR module in restoring high-frequency details. From the functional perspective, MFD mainly serves as the architectural prerequisite for frequency-domain processing by decomposing spatially entangled cross-modal features into independent low- and high-frequency subbands. Based on this decomposition, TLA enhances low-frequency thermal localization using infrared local contrast, while SHR complements it by restoring high-frequency structural details in smoke-obscured regions. Overall, these results show that MFD, TLA, and SHR are designed to work as a collaborative pipeline, in which MFD provides the structural basis for frequency-domain separation, while TLA and SHR contribute the main task-specific asymmetric enhancement.

Additionally, the results on M3FD provide a preliminary observation regarding the influence of SHR on a general dual-modal detection dataset. Compared to the MFD-only setting, adding SHR improves the mAP from 55.72% to 56.38%. This indicates that SHR does not cause an overall performance drop on M3FD. However, this is an average observation across the entire dataset. It cannot fully validate the behavior of SHR in the ambiguous non-smoke dark regions discussed earlier. More targeted data are needed to study this specific issue further.

Table 5. Ablation study of each component on the M3FD dataset.

MFD	TLA	SHR	mAP50	mAP75	mAP
-	-	-	80.94	59.67	55.34
✓	-	-	81.42	60.37	55.72
✓	✓	-	84.27	62.92	58.31
✓	-	✓	81.88	61.04	56.38
✓	✓	✓	84.79	63.28	58.62

(4) Selection of Low-Frequency Guidance Strategies. Table 6 compares the performance of visible-guided and infrared-guided low-frequency aggregation strategies. The infrared-guided strategy achieved mAP50, mAP75, and mAP of 96.67%, 74.90%, and 66.94%, respectively, outperforming the visible-guided strategy across all metrics (96.21%, 73.70%, and 66.32%, respectively). These results directly confirm the core physical assumption of this paper at the experimental level: in the low-frequency subbands of fire scenarios, infrared thermal radiation signals have a natural advantage for target localization, and an infrared-guided asymmetric aggregation strategy is therefore a correct and necessary design choice.

Table 6. Guidance selection for low-frequency components.

Methods	mAP50	mAP75	mAP
Visible guidance	96.21	73.70	66.32
Infrared guidance	96.67	74.90	66.94

(5) Validation of Multimodal Input Effectiveness. Table 7 validates the necessity of dual-modal input by comparing different input combinations. The single-modal YOLO11 baseline achieved mAP scores of 60.23% and 59.65% under visible and infrared inputs, respectively. When AFDNet used same-modality dual input (VIS + VIS or IR + IR), the mAP reached 61.31% and 61.01%, respectively, which was only a slight improvement over the single-modal baseline. This indicates that the performance gain does not come from the dual-stream architecture itself, but from the effective use of complementary cross-modal information. Under true cross-modal input (VIS + IR), AFDNet achieved an mAP of 66.94%, improving the single-modal baseline by more than 6 percentage points. This confirms that the asymmetric complementary fusion of visible and infrared modalities in the frequency domain is the fundamental source of AFDNet's performance gains.

Table 7. Comparison of different inputs for YOLO11 and AFDNet.

Methods	Input	mAP50	mAP75	mAP
YOLO11	VIS	93.74	64.33	60.23
	IR	92.13	64.37	59.65
AFDNet (ours)	VIS + VIS	94.51	65.42	61.31
	IR + IR	93.36	65.44	61.01
	VIS + IR	96.67	74.90	66.94

(6) Efficiency Comparison. Table 8 compares AFDNet with three single-stage methods in terms of parameter count, computational complexity, and inference speed. AFDNet has 5.51 million parameters, the lowest among all compared methods, and a computational complexity of 12.99 GFLOPs, which falls in the middle range. AFDNet achieved an inference speed of 123.46 FPS on an NVIDIA RTX 4090 GPU, showing strong real-time potential under server-grade hardware conditions. Compared to EI^2Det [13], AFDNet achieved better detection accuracy while reducing the parameter count by 59.2%, showing a strong balance between performance and efficiency. These results provide a promising performance foundation. However, direct deployment on resource-constrained drone platforms will require further optimization to accommodate significant differences in power budgets and computational capacity.

Table 8. Efficiency comparison of different methods.

Methods	Parameters (M)	FLOPs (G)	FPS
EI^2Det [13]	13.50	23.80	204.00
ICAFusion [43]	5.92	7.82	454.00
CFT [42]	11.30	9.32	133.30
AFDNet (ours)	5.51	12.99	123.46

4.4. Qualitative Experiments

To qualitatively evaluate the model's feature responses and detection performance, this paper presents two sets of visualization results. Figure 4 compares the Grad-CAM feature attention heatmaps of the baseline model and AFDNet in two typical smoke-obscured scenarios. The baseline model's activation regions are mainly concentrated in the bright, prominent areas of the thermal infrared image, with a relatively weak response to smoke-obscured regions. In contrast, AFDNet, after introducing frequency-separated processing, shows more focused activation in the target areas and also produces effective feature responses in smoke-obscured regions. This suggests that the combination of low-frequency infrared guidance and high-frequency smoke perception is reflected at the feature activation level. Figure 5 further compares the detection performance of AFDNet against three other methods: U2Fusion [63], DCEvo [32], and CFT [42], across several typical fire scenarios. When facing degraded visual conditions such as dense smoke, some of the compared methods tend to produce false positives and show low confidence scores. In contrast, AFDNet uses frequency-domain separation to reduce smoke interference to some extent and localize targets more accurately.

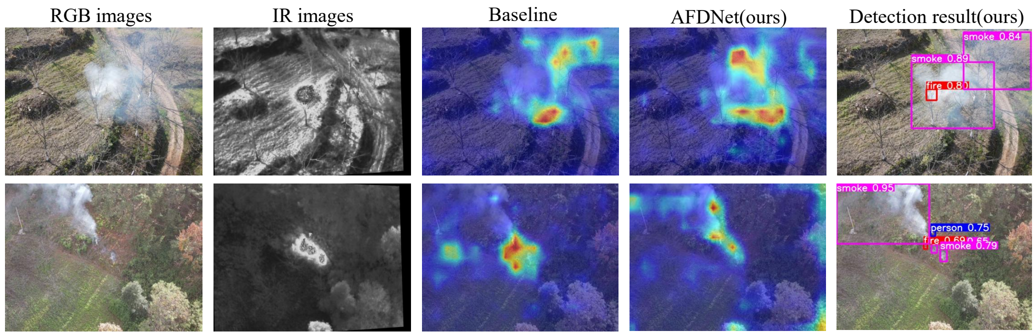

Figure 4. Comparison of feature attention heatmaps between the baseline and AFDNet.

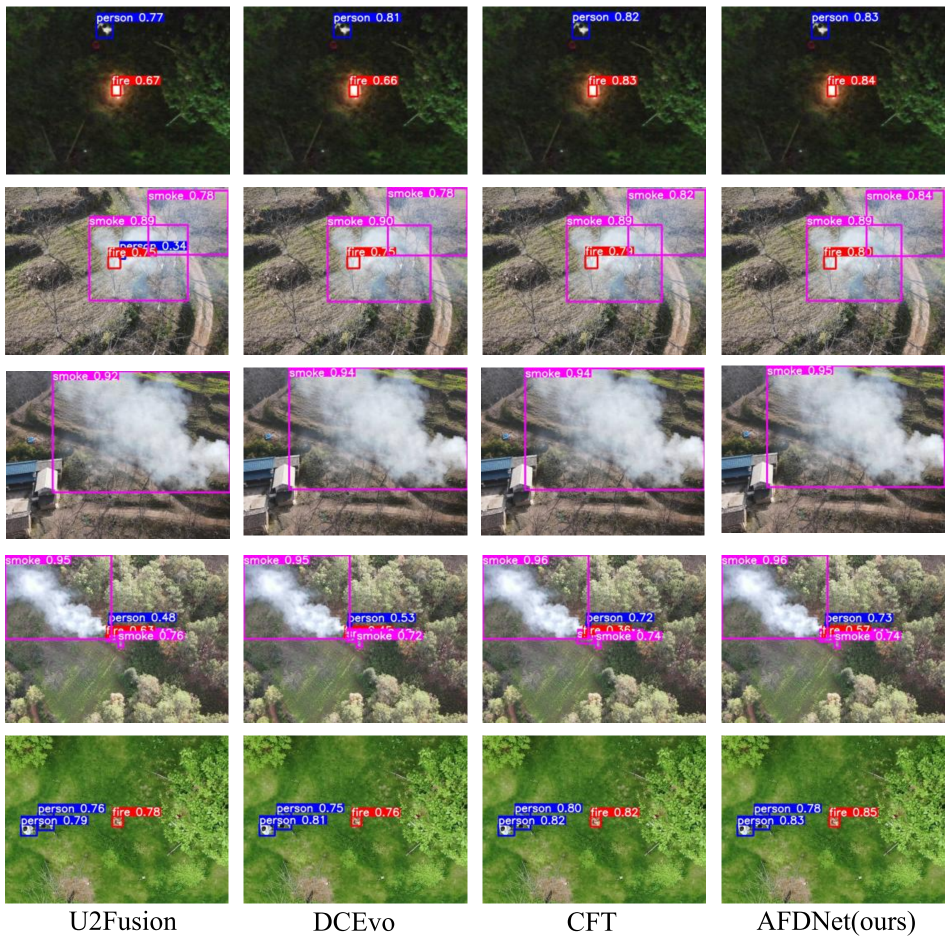

Figure 5. Visual comparison of detection performance among different methods on the RGBT-3M dataset.

5. Conclusions

This study critically addresses the pervasive bottleneck of cross-modal contamination inherent in conventional spatially symmetric fusion paradigms, particularly within complex wildfire detection scenarios. By shifting the perspective to the frequency domain, we uncover a profound physical asymmetry between the visible and infrared modalities. Driven by this insight, we propose the Asymmetric Frequency-Decoupled Network (AFDNet). Rather than enforcing full-band spatial integration, AFDNet disentangles cross-modal conflicts into distinct frequency sub-bands, deploying tailored asymmetric interaction strategies. Specifically, it adaptively leverages infrared thermal cues in low-frequency components for target localization while incorporating visible semantics for auxiliary discrimination, and uses visible high-frequency degradation as an occlusion-related reliability cue to introduce infrared details for supplementing smoke-weakened structural information. Extensive evaluations on the challenging RGBT-3M wildfire dataset and the general-purpose M3FD benchmark demonstrate that AFDNet not only achieves state-of-the-art detection accuracy under severe smoke occlusion and variable lighting but also exhibits promising cross-scenario generalization.

However, this study has certain limitations. Current datasets lack sufficient examples of extreme environmental conditions. These conditions include high ambient temperatures, low thermal contrast, dense vegetation shadows, and dark skies. Consequently, the robustness of the TLA and SHR modules under these boundary cases requires further verification. For instance, high background temperatures can reduce thermal contrast and affect the reliability weight in the TLA module. Furthermore, non-smoke dark regions in the visible spectrum might trigger ambiguous responses in the SHR module. While our current experiments on the available datasets do not show an overall negative impact, more targeted data are necessary to fully evaluate these special cases.

Moving forward, future research will extend this asymmetric frequency-domain approach to more extreme environmental conditions. We also plan to collect more diverse visible–infrared wildfire data to evaluate these boundary cases. Finally, we will optimize the network architecture for lightweight and real-time deployment on resource-constrained edge computing devices.

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Data Availability Statement: The RGBT-3M dataset presented in this study is publicly accessible at https://complex.ustc.edu.cn/sjwwataset/list.htm, accessed on 6 April 2026. The M3FD dataset can be accessed via https://github.com/dlut-dimt/TarDAL, accessed on 6 April 2026.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Quantitative Validation of Frequency-Domain Asymmetry

This paper proposes a frequency-domain asymmetry hypothesis with two core claims. First, in local regions containing thermal targets, the infrared modality has significantly higher local contrast in the low-frequency subband. Second, in smoke-occluded regions, the visible high-frequency subband undergoes energy collapse, which serves as a natural occlusion prior. Global energy statistics can produce misleading results due to the dominance of background pixels. To avoid this, we design two quantitative experiments based on annotated bounding boxes from the RGBT-3M validation set, using strictly paired foreground and background patches.

All patches are normalized before applying 2D Discrete Wavelet Transform with a Haar wavelet basis. Two metrics are evaluated. The first is the Local Contrast Ratio (LCR), defined as the ratio of the LL subband variance in the target region to that in the background region. A value greater than 1 indicates that the target is more salient than the background. The second metric is the absolute high-frequency energy (variance of the high-frequency subbands). We directly compare this energy between smoke-occluded regions and adjacent clear non-smoke regions to explicitly verify the response collapse.

Appendix A.1. Low-Frequency Local Contrast Analysis for Thermal Targets

Figure A1 shows the LCR comparison results. This analysis uses 3402 paired samples of flame and people targets from the RGBT-3M validation set. The median LCR of the infrared modality is 55.59, while that of the visible modality is only 3.16. This represents a difference of approximately 17.6 times (Wilcoxon signed-rank test, $p < 0.001$). This result quantitatively confirms that thermal radiation energy forms a highly concentrated peak response in the infrared low-frequency subband. Its local saliency is far greater than that of the visible modality. This provides direct empirical support for the infrared-dominant asymmetric aggregation strategy used in the TLA module.

The physical origin of this result lies in the large difference in radiation intensity between thermal targets and the background. Flames and human bodies have very high radiant exitance in the infrared band, while background vegetation and soil have relatively uniform and low radiation intensity. This strong radiation contrast appears in the infrared low-frequency subband as a sharp energy peak in the target region. This produces a local contrast that is far higher than that of the visible modality [66].

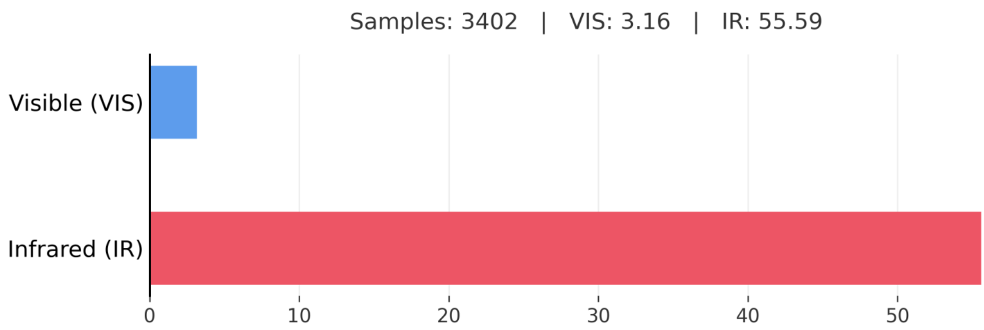

Figure A1. Comparison of Low-Frequency Local Contrast Ratio (LCR) for Thermal Target Regions (RGBT-3M validation set, flame and people categories, 3402 paired samples).

Appendix A.2. Visible High-Frequency Energy Collapse in Smoke Regions

Figure A2 compares the absolute high-frequency energy responses between smoke-occluded regions and clear non-smoke regions in the visible modality. This analysis uses 3481 paired smoke samples from the RGBT-3M validation set. Instead of using global statistics, we extract the absolute high-frequency energy of the foreground smoke and compare it directly to the adjacent clear background.

The absolute values strongly validate the response collapse hypothesis. In the visible modality, the mean high-frequency energy of smoke-occluded regions drops drastically to 0.003581, compared to 0.006554 in clear background regions. This direct reduction quantitatively confirms the severe texture loss caused by smoke scattering. This statistical confirmation of visible energy collapse effectively underpins the design logic of the proposed SHR module. The module creatively leverages these collapsed regions as a natural occlusion mask.

This result is consistent with Mie scattering theory. The strong scattering of visible light by smoke particles homogenizes local spatial gradients, causing a large drop in texture energy in smoke-covered regions [25].

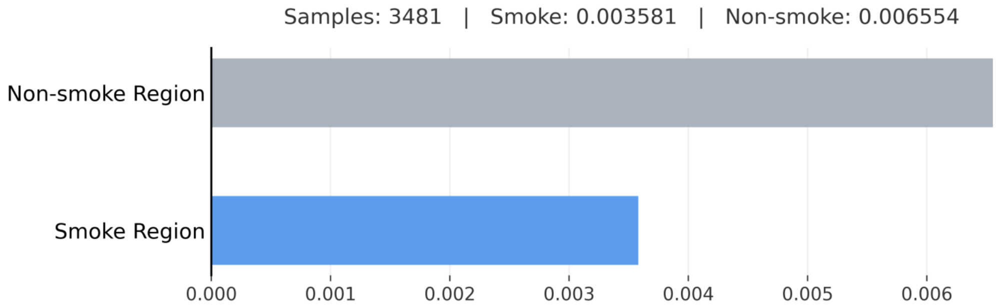

Figure A2. Comparison of absolute high-frequency energy between smoke-occluded regions and clear non-smoke regions in the visible modality (RGBT-3M validation set, smoke category, 3481 paired samples).

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